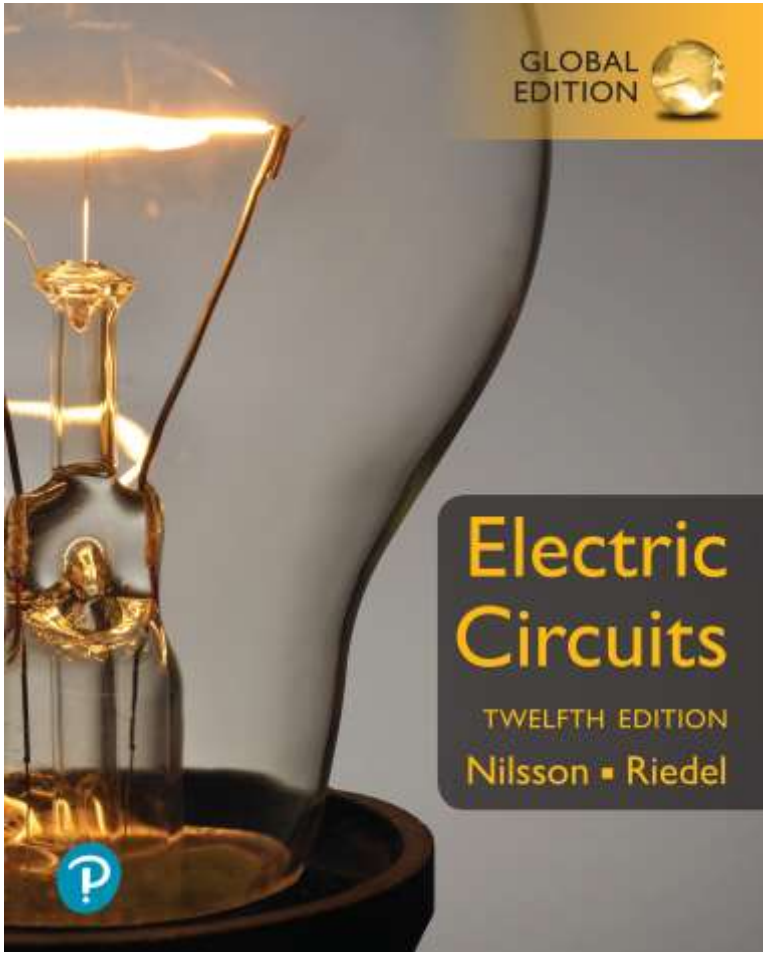
Electric Circuits

Twelfth Edition, Global Edition



Chapter 5 – Part 3

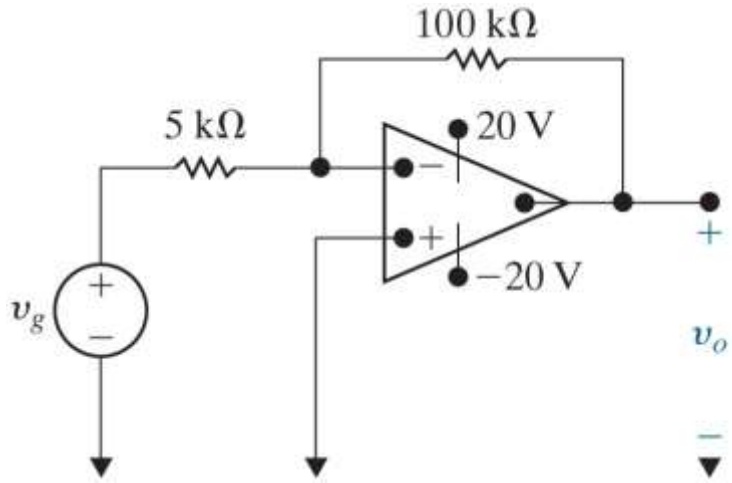
The Operational Amplifier

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Chapter 5 – Op Amps (2 of 4)

What op amp circuit configuration is shown here?



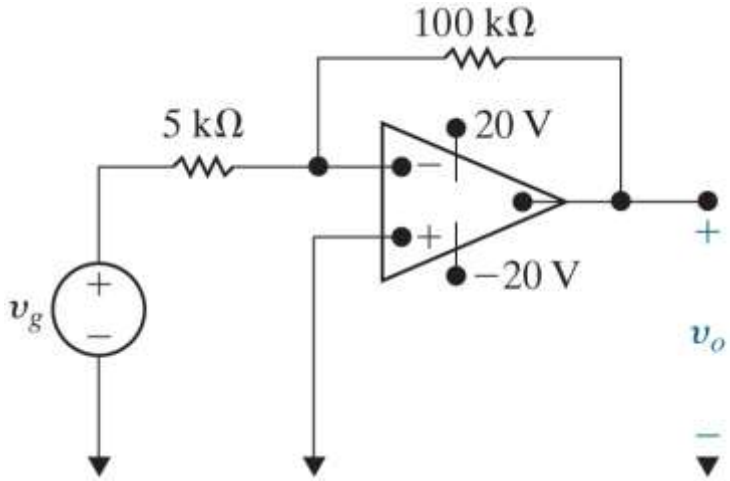
- ✔ A. Inverting amplifier
- ✘ B. Inverting summing amplifier
- ✘ C. Non-inverting amplifier
- ✘ D. Non-inverting summing amplifier

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Chapter 5 – Op Amps (3 of 4)

Suppose $v_g = 0.5\,\text{V}$. Assuming the op amp is ideal, what is v_o?



- ✗ A. 10 V
- ✗ B. −20 V
- ✓ C. − 10 V

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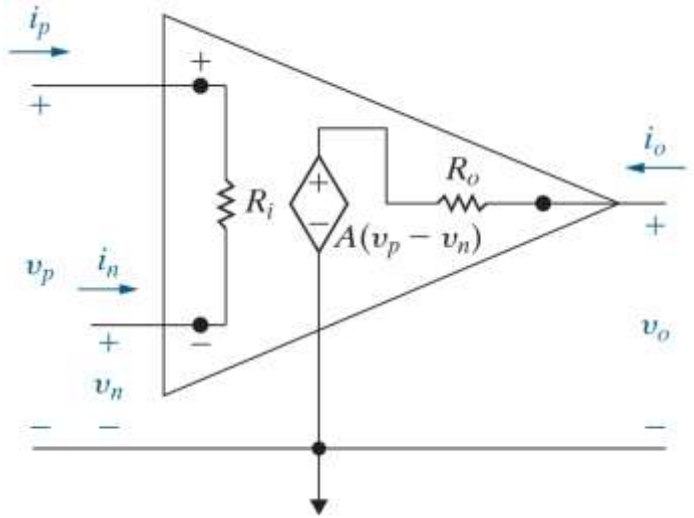
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Chapter 5 – Op Amps (1 of 4)

A more realistic op amp model

Predicts op amp behavior in its linear region and includes three modifications to the ideal op amp:

- A finite input resistance
- A finite open-loop gain
- A non-zero output resistance

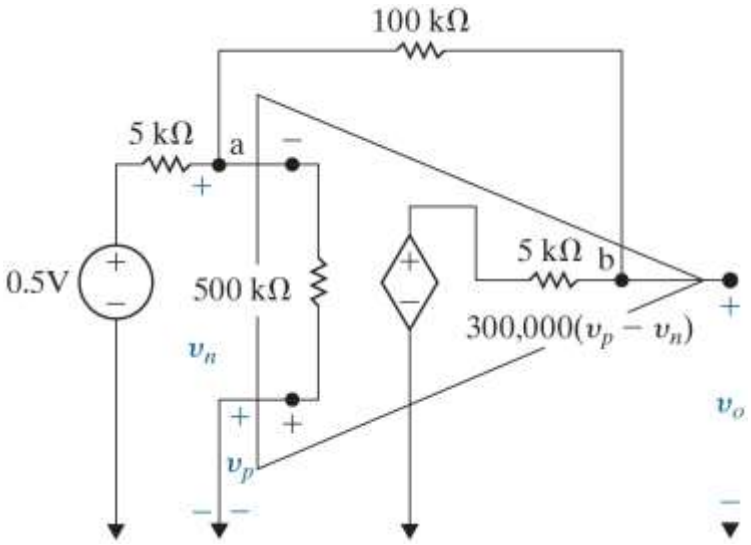


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Chapter 5 – Op Amps (4 of 4)

Now let's use the more realistic model for the op amp to analyze the inverting amplifier. Assume the input resistance is $500\ \text{k}\Omega$, the output resistance is $5\ \text{k}\Omega$, and the open-loop gain is 300,000.



Write KCL equations at nodes a and b:

node a: $\dfrac{v_n - 0.5}{5000} + \dfrac{v_n}{500{,}000} + \dfrac{v_n - v_o}{100{,}000} = 0$

node b: $\dfrac{v_o - v_n}{100{,}000} + \dfrac{v_o - 300{,}000(-v_n)}{5000} = 0$

Using a calculator to solve these equations simultaneously gives $v_o = -9.999258$ V. Therefore, the ideal op amp assumptions do a good job approximating an actual op amp.

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